Toward Ethical AI Through Bayesian Uncertainty in Neural Question Answering

Riccardo Di Sipio

Dayforce HCM

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Abstract

We explore Bayesian reasoning as a means to quantify uncertainty in neural networks for question answering. Starting with a multilayer perceptron on the Iris dataset, we show how posterior inference conveys confidence in predictions. We then extend this to language models, applying Bayesian inference first to a frozen head and finally to LoRA-adapted transformers, evaluated on the CommonsenseQA benchmark. Rather than aiming for state-of-the-art accuracy, we compare Laplace approximations against maximum a posteriori (MAP) estimates to highlight uncertainty calibration and selective prediction. This allows models to abstain when confidence is low. An "I don't know" response not only improves interpretability but also illustrates how Bayesian methods can contribute to more responsible and ethical deployment of neural question-answering systems.

1 Introduction

Large neural networks have achieved remarkable progress in natural language processing, particularly in tasks such as question answering [1–4]. Yet despite their predictive power, these models typically produce point estimates without a measure of confidence [5–8]. This absence of calibrated uncertainty can be problematic: models may answer confidently even when wrong, which is particularly concerning in high-stakes applications [9–12].

In a standard neural network, training produces a single set of parameters θ, and predictions are made as point estimates $p(y|x, \theta)$. In contrast, Bayesian reasoning treats the parameters themselves as random variables with a distribution that captures our uncertainty. A prior distribution $p(\theta)$ encodes initial beliefs, the likelihood $p(y|x, \theta)$ links data to parameters, and Bayes' rule gives a posterior $p(\theta|\mathcal{D}) \propto p(\mathcal{D}|\theta)p(\theta)$ after observing data $\mathcal{D}$. Predictions then marginalize over this posterior:

$$p(y|x, \mathcal{D}) = \int p(y|x, \theta)\, p(\theta|\mathcal{D})\, d\theta.$$

This marginalization yields predictive distributions that reflect both the data fit and the epistemic uncertainty of the model. In practice, exact posteriors are intractable for large networks, so approximations such as Laplace [13] or Monte Carlo [14, 15] sampling are employed.

Thus, Bayesian reasoning provides a principled framework for addressing the lack of calibrated uncertainties in neural networks [16]. By combining priors with likelihoods to form posteriors, Bayesian methods not only make predictions but also quantify the level of belief in those predictions. This opens the possibility for models to abstain *i.e.* to say "I don't know" when confidence is low. Such behavior is not only useful for downstream tasks like selective prediction and calibration, but also central to building systems that align with the goals of responsible and ethical AI [12, 17–19].

From an ethical standpoint, the ability of a model to express calibrated uncertainty embodies a form of *epistemic humility*, a recognition of the limits of its own knowledge. In safety- or policy-critical contexts, such behaviour fosters accountability and trust: an uncertain system can defer to human judgment rather than propagate confident errors. Within this framework, Bayesian inference provides not only probabilistic estimates but also a normative stance, preferring caution over overconfidence.

Despite extensive research on uncertainty calibration and robustness in neural networks (*e.g.* [7, 8, 20–22]), little work has examined how Bayesian inference can be operationalized in natural language question answering to support responsible AI behaviour. Prior studies typically use uncertainty estimates to improve model confidence scores [8, 22, 23] or ensemble diversity and robustness under distributional shift [20, 24]), rather than to enable systems that can recognize and communicate when their knowledge is insufficient. This study addresses that gap by demonstrating, through three progressively complex experiments, how Bayesian reasoning can endow neural QA models with the ability to express calibrated uncertainty, supporting ethically aligned behaviour such as abstention or transparency in low-confidence scenarios. Unlike prior work that treats uncertainty estimation primarily as a technical calibration problem, we investigate it as a foundation for ethically grounded model behaviour. Our goal is to show, through controlled experimental progression, that Bayesian posteriors can serve as actionable confidence signals, bridging statistical inference with responsible system design.

In this paper, we explore this idea through three experiments of increasing complexity, each chosen to isolate a distinct aspect of the problem, from theoretical transparency to practical scalability:

1. **Bayesian inference in a controlled setting (Iris dataset):** We revisit a multilayer perceptron trained with Hamiltonian Monte Carlo sampling to visualize how posterior predictive distributions naturally express epistemic uncertainty.

2. **Bayesian head on a frozen transformer:** We transfer these insights to language data by training a Bayesian linear head on top of a frozen BERT model, illustrating how posterior uncertainty translates to selective prediction in a simple QA setup.

3. **LoRA fine-tuning with Laplace approximation:** Finally, we apply a Laplace approximation over adapter and head parameters of a LoRA-tuned BERT-base model, evaluated on CommonsenseQA [25]. This setup demonstrates that uncertainty-aware transformers can be realized efficiently without retraining the full model.

Across these experiments, we show that Bayesian posteriors can improve calibration, enable interpretable abstention, and provide principled confidence estimates. The contribution of this work lies in illustrating how classical Bayesian tools such as MCMC sampling and Laplace approximations can be operationalized in modern language models to support ethical AI principles such as transparency, caution, and user trust.

2 Related Work

Question Answering Benchmarks. Commonsense reasoning has become a standard testbed for evaluating language models beyond surface-level pattern matching. The CommonsenseQA dataset introduced by Talmor et al. [25] provides multiple-choice questions designed to probe background knowledge and reasoning ability, and remains a widely used benchmark for evaluating uncertainty and reasoning in neural question answering.

Parameter-Efficient Fine-Tuning. Transformer-based models such as BERT [1] have motivated methods for efficient adaptation to downstream tasks. LoRA (Low-Rank Adaptation) introduced by Hu et al. [26] injects trainable low-rank matrices into frozen weights, reducing memory and compute while retaining performance. LoRA has since been applied broadly in large language models, including in uncertainty-aware fine-tuning [27, 28]. These methods enable selective adaptation of small subsets of parameters, which we leverage in Experiment 3 to make Bayesian posterior estimation computationally tractable.

Bayesian Neural Networks and Uncertainty Estimation. Bayesian inference for neural networks has a long history, including exact posterior sampling via Markov Chain Monte Carlo (MCMC) [14], variational inference, and Laplace approximations [13]. Modern extensions have demonstrated the scalability of Laplace approximations for deep models [21, 29], and uncertainty estimation via dropout [7], ensembles [20, 23], or evidential inference [22]. These approaches primarily aim to improve predictive calibration and robustness under distributional shift [8, 24], treating uncertainty as a statistical property of the model rather than an interpretable signal for user-facing decision support.

Uncertainty in Natural Language Processing. A growing body of work explores Bayesian and calibration methods in NLP, including for text classification and question answering [30, 31]. Reliability diagrams, selective prediction, and abstention mechanisms have been studied as ways to align model confidence with empirical accuracy [5]. However, prior NLP research has largely focused on improving performance metrics such as calibration error or accuracy under shift, without explicitly linking uncertainty estimation to ethically grounded system behaviour.

Research Gap. Across these threads, uncertainty is typically used to adjust confidence scores, improve ensemble diversity, or regularize learning. What remains underexplored is how Bayesian reasoning can be operationalized in question answering to enable *transparent abstention and uncertainty communication*, *i.e.* models capable of signalling when their knowledge is insufficient. Our work addresses this gap by demonstrating, through a sequence of experiments of increasing realism, how classical Bayesian inference methods (MCMC and Laplace) can enrich transformer-based QA with calibrated uncertainty estimates that support ethical principles of transparency and trust.

3 Ethical Implications of Uncertainty Communication

Bayesian inference provides not only a statistical framework for reasoning under uncertainty but also a normative one for representing the limits of an intelligent system's knowledge. From an ethical standpoint, this capacity connects directly to principles of transparency, accountability, and human oversight identified in contemporary AI ethics frameworks [17–19]. A model that expresses confidence distributions rather than unqualified predictions enables value-aligned decision-making, where uncertainty itself becomes an instrument of responsible behaviour.

Most prior studies on Bayesian uncertainty in neural models have concentrated on predictive calibration and robustness rather than on ethical or decision-theoretic implications. This emphasis stems from the origins of Bayesian deep learning in reliability engineering and uncertainty quantification, where the goal was to improve model confidence estimates and mitigate overfitting under distributional shift. For instance, Monte Carlo dropout [7] treats stochastic regularization as approximate Bayesian inference, improving calibration but offering no mechanism for how those confidence values should guide user decisions. Deep ensembles [20] achieve strong empirical uncertainty estimates by averaging multiple networks, yet they remain a purely technical ensemble method without interpretive communication of uncertainty to end users. Similarly, Bayesian last-layer approaches such as the Laplace approximation of Ritter et al. [21] and related extensions [29] provide tractable posterior estimates for neural classifiers, but these probabilities are typically evaluated only as internal performance metrics (e.g., Brier score, ECE) rather than as communicative signals in human-AI interaction. Across these lines of research, uncertainty serves an operational purpose, making predictions numerically reliable, but remains internal to the model and disconnected from ethical considerations about its use.

By contrast, our focus lies on the *external* role of uncertainty: how probabilistic confidence can be communicated or acted upon in ways that reflect human norms of caution, responsibility, and accountability. This perspective introduces an analytical distinction between *calibration* and *communication*: a model that merely produces well-calibrated probabilities may still behave unethically if it presents those probabilities without context or qualification. We therefore reinterpret Bayesian reasoning not only as a statistical safeguard but also as a mechanism for ethically aligned behaviour in AI systems.

While the ability to abstain is one ethically grounded outcome of uncertainty quantification, it is not the only one. In some domains such as education or scientific research, withholding an answer may be appropriate and even desirable, signalling the limits of the model's knowledge, a stance often described in the literature as *epistemic humility*. In others, particularly decision-support contexts where a choice must ultimately be made, explicitly *flagging* low-confidence outputs may be more valuable than full abstention. Rather than refusing to answer, the model can communicate its uncertainty (*e.g.* , "this is a low-confidence estimate") so that human users can weigh the information accordingly. Both behaviours derive from the same Bayesian principle: uncertainty should not be hidden but made *legible* to human decision-makers. This legibility transforms uncertainty from a mere calibration statistic into a communicative act that promotes justified trust, transparency, and safer use of model outputs.

In this light, Bayesian uncertainty is not simply a numerical property of a classifier but an intrinsic feature of model design. It defines how AI systems can convey epistemic responsibility to users and institutions, linking statistical reasoning with moral accountability.

4 Experimental Demonstrations

To make the discussion concrete, we present three experimental demonstrations of increasing complexity. Each experiment tests the central hypothesis that Bayesian inference enables neural question-answering models to express calibrated uncertainty that can be ethically interpreted as justified confidence or abstention. We progressively move from a simple pedagogical baseline to modern language models, holding evaluation metrics constant while varying model capacity and inference method. Across these setups, we assess how Bayesian treatments through full posterior sampling or Laplace approximation affect predictive calibration, selective prediction, and interpretability. The emphasis is educational rather than competitive: our goal is to highlight transparency, reliability, and ethical deployment rather than state-of-the-art accuracy. All experiments are evaluated using accuracy, Brier score, Expected Calibration Error (ECE), and accuracy–coverage curves, ensuring a statistically sound comparison between MAP and Bayesian inference.

4.1 Methodological Controls and Evaluation Metrics

Across all experiments, we control for data scale and model complexity to isolate the effect of Bayesian inference on uncertainty estimation. Each setup compares a deterministic maximum-a-posteriori (MAP) baseline with a Bayesian variant using identical architectures, training data, and optimization schedules. This design ensures that observed differences can be attributed to the inference method rather than model capacity or dataset variability.

Performance is assessed using complementary metrics that jointly capture accuracy and reliability: **accuracy** (overall correctness), **Brier score** (mean squared calibration error), **Expected Calibration Error (ECE)** (bin-based deviation between confidence and accuracy), and **accuracy–coverage curves** (selective prediction behaviour under abstention thresholds). Together, these metrics provide a comprehensive view of predictive soundness and uncertainty calibration.

Full implementation code is available at [GitHub URL] (to be added post-review)

4.2 Experiment 1: Bayesian Inference on the Iris Dataset

As a pedagogical starting point, we revisit Bayesian inference on the classic Iris dataset [32], demonstrating how posterior predictive distributions reflect uncertainty in classification.

This small, well-structured dataset remains a useful teaching example, consisting of 150 labeled samples across three flower species, each described by four continuous features.

We train a simple multilayer perceptron (MLP) classifier on this dataset, but unlike standard training where weights are optimized to point estimates, we treat the weights as random variables with prior distributions. Using Markov Chain Monte Carlo (MCMC) sampling [14, 15], we draw from the posterior distribution of the weights conditioned on the data.

This Bayesian treatment allows us to:

- Visualize priors vs posteriors: showing how the data updates our initial beliefs about parameters.

- Obtain predictive distributions: instead of producing a single probability vector, the network integrates over weight samples to generate a distribution over predictions.
- Quantify uncertainty: for each test example, we can report not only the predicted class but also the posterior variance, highlighting cases where the model is unsure.

To make these ideas concrete, we reproduce three kinds of results:

1. Prior vs posterior plots for selected weights, showing how uncertainty narrows after conditioning on data.
2. Two-dimensional marginal posterior distributions for weight pairs, which sometimes exhibit correlation or multimodality.
3. Per-example posterior predictive distributions, where mean predictions are accompanied by error bars. In particular, these plots illustrate cases where one class is most probable but another remains a close runner-up, motivating the idea of an "I don't know" response when confidence is too low.

Rather than aiming for performance improvements on this simple dataset, our goal here is educational: to demonstrate how Bayesian reasoning naturally introduces the concept of belief and uncertainty, setting the stage for more complex models in subsequent experiments.

To build intuition for the Bayesian treatment of neural networks, we begin with controlled toy settings where the posterior can be visualized directly. These demonstrations illustrate how priors, likelihoods, and posteriors interact to shape the model's predictions and associated uncertainty.

At the one–dimensional level, we can compare prior and posterior distributions over individual parameters of a small network. As shown in Fig. 1, the prior is broad and uninformative, while the posterior becomes more concentrated around values supported by the data. This highlights how Bayesian updating reduces uncertainty when evidence accumulates.

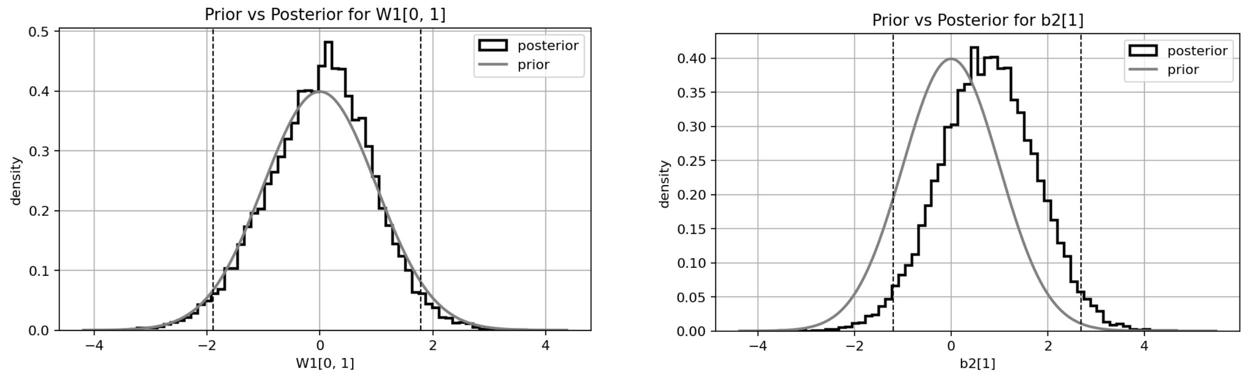
In higher dimensions, posteriors capture not only marginal variance but also correlations between parameters. Fig. 2 contrasts two examples: one where parameters remain nearly independent, and another where strong posterior correlation emerges. This illustrates how the geometry of the parameter space is reshaped by data, an effect that is invisible in maximum–likelihood or MAP estimates.

The effect of this parameter uncertainty propagates naturally to predictions. Fig. 3 shows per–question posterior predictive distributions for individual examples, including the mean probability, one–sigma error bars, the predicted class, and the true label. These plots make explicit when the model is confidently correct, confidently wrong, or uncertain.

Finally, we examine how posterior predictive uncertainty translates into better calibration and more cautious decision making. Fig. 4 presents two complementary diagnostics. A reliability diagram compares predicted confidence against empirical accuracy, in which perfect calibration corresponds to the diagonal line, while an accuracy–coverage curve evaluates selective prediction by plotting accuracy on answered cases as a function of coverage, *i.e.* , the fraction of questions the model elects to answer when abstaining below a confidence threshold. Together, these metrics show that the Bayesian treatment mitigates overconfidence and enables a principled trade-off between reliability and coverage.

In practical deployment, a confidence threshold τ governs whether the model answers or abstains. Varying τ traces the accuracy–coverage frontier (again in Fig. 4), revealing how uncertainty quantification can support ethically grounded system design. Here, the *conditional accuracy* at a given threshold, *i.e.* the accuracy computed only on the subset of predictions with confidence above τ, quantifies how reliable the model is when it chooses to answer. Although no single threshold generalizes across applications, τ can be linked empirically to desired reliability levels: for instance, a system designer may choose τ such that the conditional accuracy of accepted predictions exceeds a target (e.g., 90%). This calibration-based criterion transforms τ

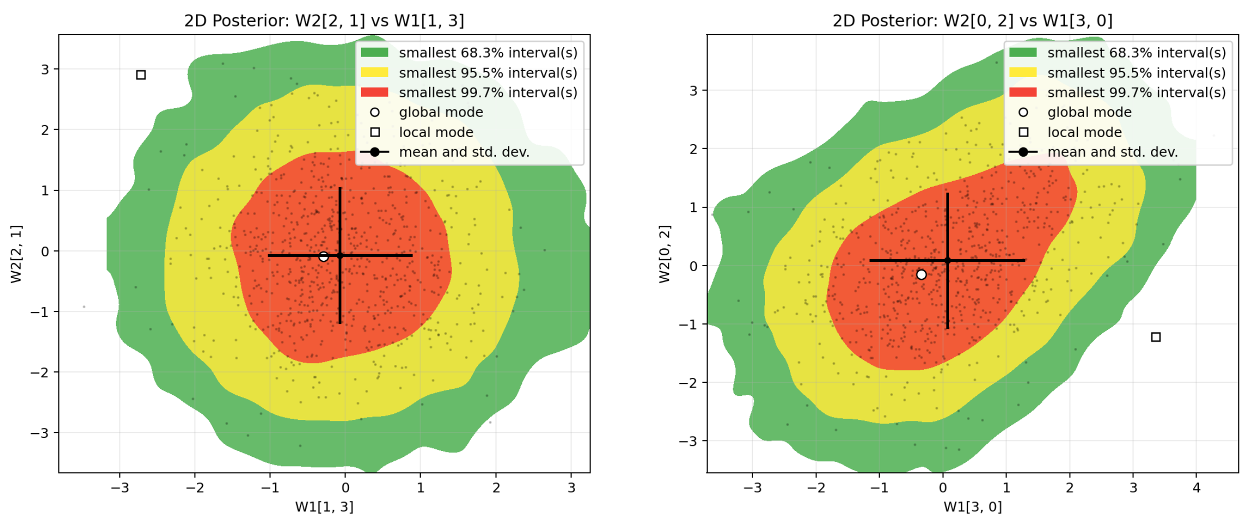
into a measurable policy parameter rather than an arbitrary cutoff. In high-stakes applications, selecting a higher τ encodes a normative preference for transparency and abstention over unwarranted certainty, ensuring that the model only responds when its expected reliability meets human-defined standards.



(a) Prior vs. posterior for $W_1[0,1]$.

(b) Prior vs. posterior for $b_2[1]$.

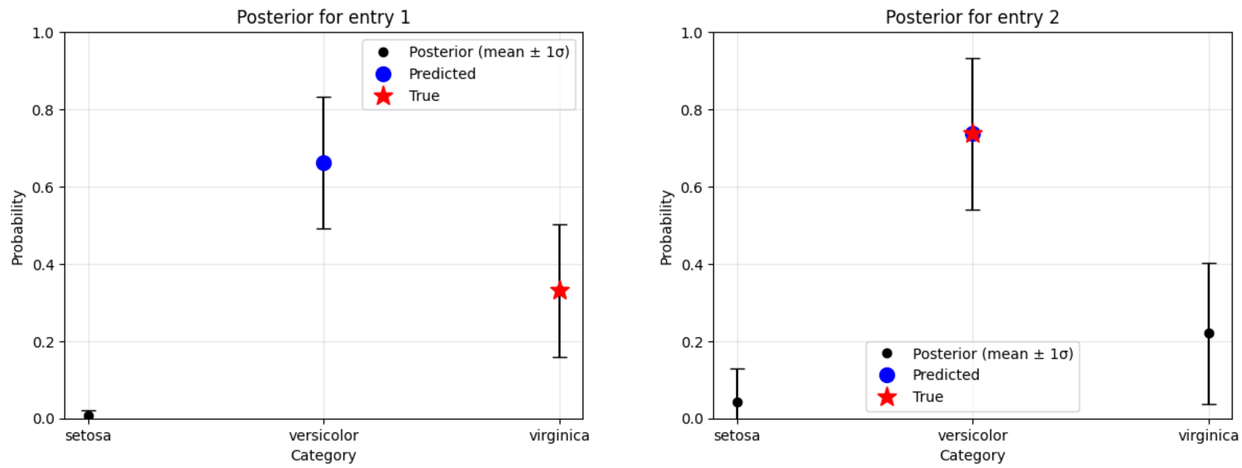
Figure 1: One–dimensional priors (gray) and marginalized posteriors (black) for selected parameters. Posteriors concentrate and shift relative to priors as the data updates beliefs.



(a) $W_1[1,3]$ vs. $W_2[2,1]$: near-circular (weak correlation).

(b) $W_1[3,0]$ vs. $W_2[0,2]$: tilted contours (correlated).

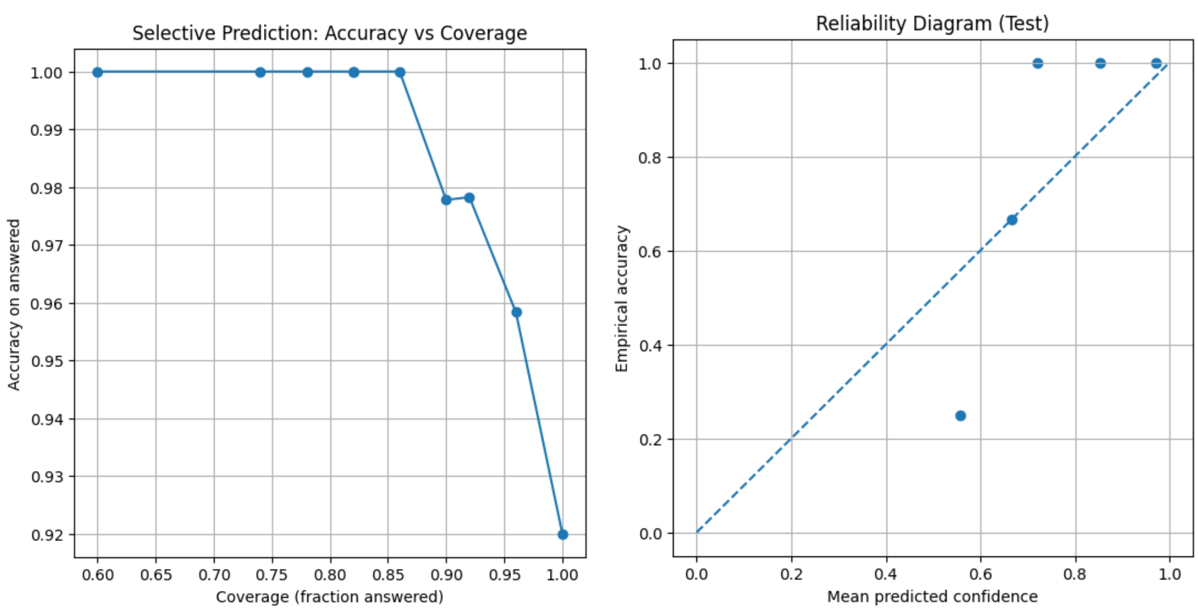
Figure 2: Two–dimensional marginalized posteriors with credible-region contours (68/95/99.7%). Geometry reveals how uncertainty couples parameters.



(a) Entry #1: confident correct prediction.

(b) Entry #2: higher uncertainty (wider error bars).

Figure 3: Posterior predictive per sample (mean $\pm$ 1σ). Black dots = means; blue circle = predicted class; red star = true class.



(a) Selective prediction: accuracy vs. coverage.

(b) Reliability diagram (test).

Figure 4: **Calibration and selective prediction.** (*Left*) Accuracy–coverage curve showing the trade-off between reliability and coverage as the confidence threshold τ varies. The Bayesian posterior better aligns confidence with correctness and enables abstention when uncertainty is high. (*Right*) Reliability diagram comparing predicted confidence with empirical accuracy; perfect calibration lies along the diagonal.

4.3 Experiment 2: A small-scale question answering task with BERT

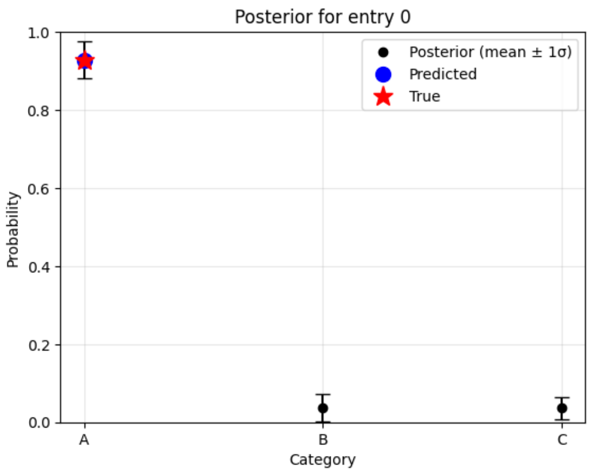
To bridge from toy problems to real benchmarks, we constructed a small synthetic question answering dataset with three answer options per question (examples listed in Appendix). Questions span general knowledge domains such as geography, history, and basic science, with one correct option and two distractors.

We use `bert-large-uncased` [1] to encode each (question, option) pair, extract the `[CLS]` embedding, and concatenate the resulting three vectors into a single feature representation. This representation is then passed to a Bayesian multinomial logistic regression head, trained using Hamiltonian Monte Carlo (HMC) with the NUTS sampler [33] as implemented in NumPyro [34]. For this experiment, we used the `large` rather than the smaller `base` variant to ensure sufficiently rich contextual representations, since this setup does not involve any fine-tuning of the transformer backbone. The larger model provides stronger pretrained features for the Bayesian head to operate on, allowing the analysis to focus on the effect of uncertainty modeling rather than limitations of the base encoder.

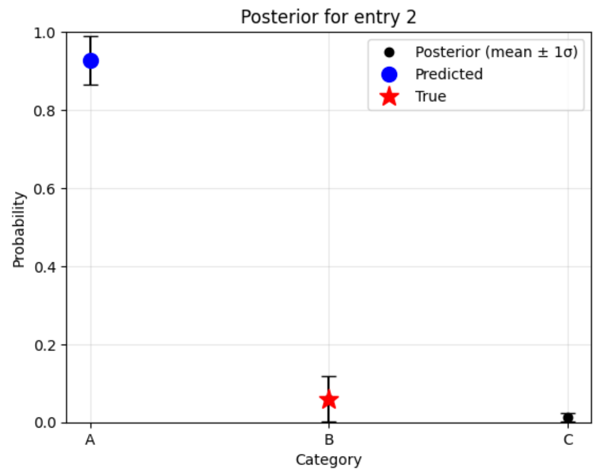
Because the feature dimension is modest (1024 embedding dimensions), full posterior sampling is computationally feasible, allowing us to characterize epistemic uncertainty directly rather than through approximations. This setup highlights how a Bayesian treatment can be layered on top of a pretrained encoder, even when resources are limited.

Figure 5 shows posterior distributions for individual entries in the toy QA dataset. In Entry 0, the model is confident and correct: the predicted answer aligns with the ground truth, and the posterior variance is small. In Entry 3, uncertainty is higher, and the posterior reflects ambiguity between two plausible answers. By contrast, Entry 2 (shown in Fig. 5b) illustrates a failure case: the model is highly confident but incorrect. This highlights the importance of evaluating not only accuracy but also calibration and coverage, since confidence alone may mislead users when the model is wrong.

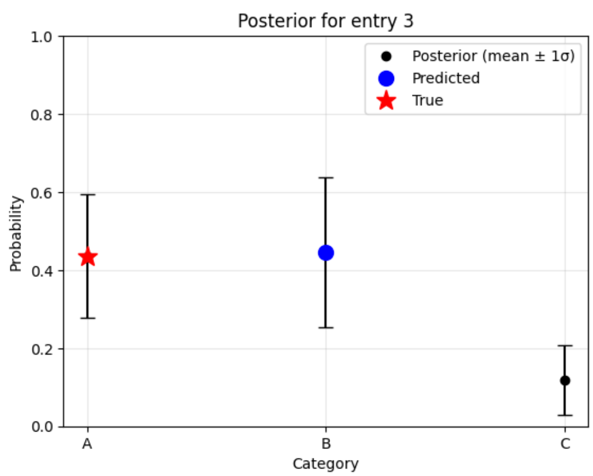
Beyond single–question posteriors, we also examined aggregate calibration on the toy QA dataset. Fig. 6 (left) shows the accuracy–coverage curve: as the confidence threshold increases, the model abstains on more examples, yielding higher accuracy on the subset it does answer. This indicates that the model's predicted probabilities do carry useful information about uncertainty, even if imperfect. The reliability diagram in Fig. 6 (right) provides a complementary view by comparing predicted confidence against empirical accuracy. Although the small dataset and limited training lead to noisy estimates, the plot reveals a tendency toward overconfidence, where predicted probabilities exceed the actual likelihood of correctness. These calibration artifacts foreshadow the importance of more principled Bayesian approaches explored in the following section.



(a) Posterior for entry 0

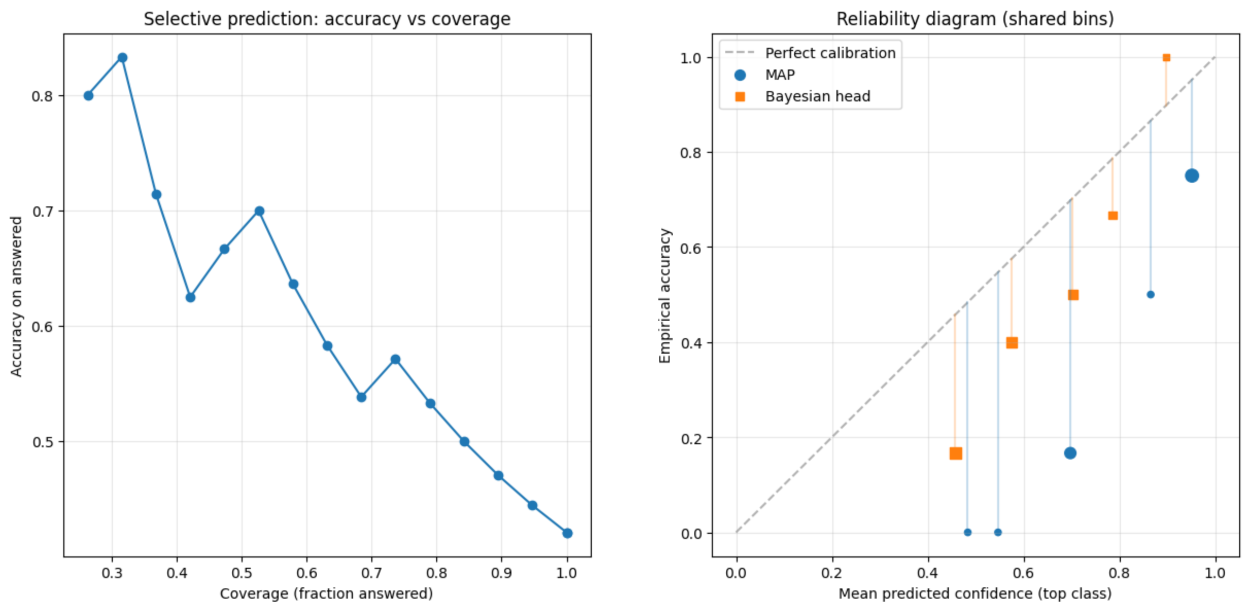


(b) Posterior for entry 2



(c) Posterior for entry 3

Figure 5: Posterior predictive distributions for selected entries in the custom three-class question answering dataset. Black points and error bars show the mean and $\pm 1\sigma$ uncertainty across posterior samples, the blue dot marks the predicted class, and the red star denotes the true label.



(a) Selective prediction

(b) Reliability diagram

Figure 6: Calibration plots for Experiment 2 with `bert-large-uncased`. (a) Accuracy–coverage trade-off, showing how performance improves when abstaining on low-confidence answers. (b) Reliability diagram, comparing mean predicted confidence with empirical accuracy across bins.

4.4 Experiment 3: LoRA with Laplace Approximation on CommonsenseQA

We scale the Bayesian treatment to a realistic QA benchmark by fine-tuning a pretrained encoder with LoRA adapters and then placing a Bayesian posterior over the small classification head.

We fine-tune `BERT-base-uncased` [1] on CommonsenseQA[25] with LoRA adapters applied to the last two transformer layers, targeting the attention projections (query, key, value) and the output dense module. The backbone BERT weights are frozen; only the LoRA parameters and a linear classification head are updated during training.

We approximate the posterior over the head parameters using a *Laplace approximation*. Concretely, after training to a maximum a posteriori (MAP) solution, we estimate the local curvature of the loss surface via the *empirical Fisher information*. For parameters θ, the Fisher is defined as

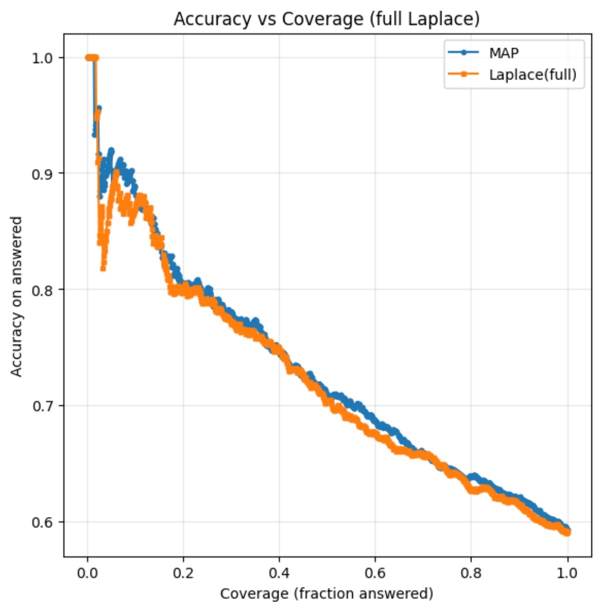
$$F(\theta) = \mathbb{E}\left[\nabla_\theta \log p(y|x,\theta)\, \nabla_\theta \log p(y|x,\theta)^\top\right],$$

which captures the sensitivity of the likelihood to changes in θ. In practice, we compute its *diagonal approximation* by averaging squared gradients over the dataset. The resulting Gaussian posterior, with mean at the MAP parameters and variance given by the inverse Fisher, provides a tractable way to sample parameter perturbations and thus quantify predictive uncertainty.

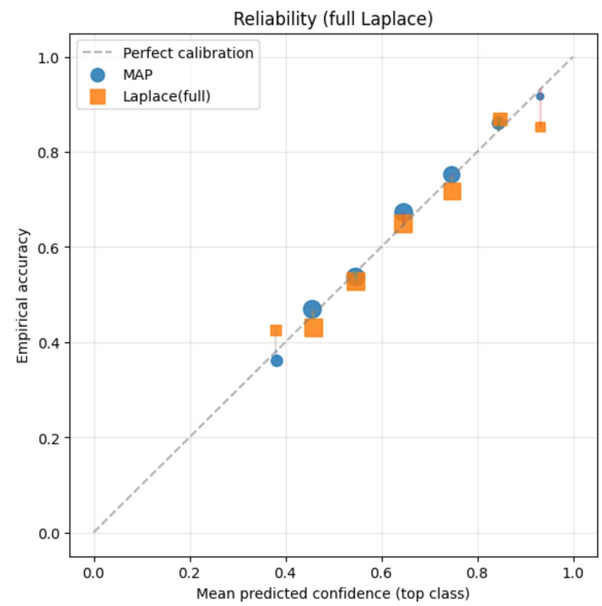
In practical terms, posterior predictive distributions are obtained by Monte Carlo sampling, typically with $S_{\mathrm{MC}} \equiv 30$, and averaging the resulting softmax outputs. This setup balances scalability with the ability to capture epistemic uncertainty in a realistic question answering setting.

For efficiency, each question's five options are reduced to three by sampling two distractors uniformly while preserving the correct answer in a random position. We optionally subsample the training set for faster runs.

The selective prediction curve in Fig. 7(a) compares the accuracy–coverage trade-off for maximum a posteriori (MAP) predictions and their Laplace-approximated counterparts. As coverage decreases (the



(a) Accuracy vs coverage



(b) Reliability diagram

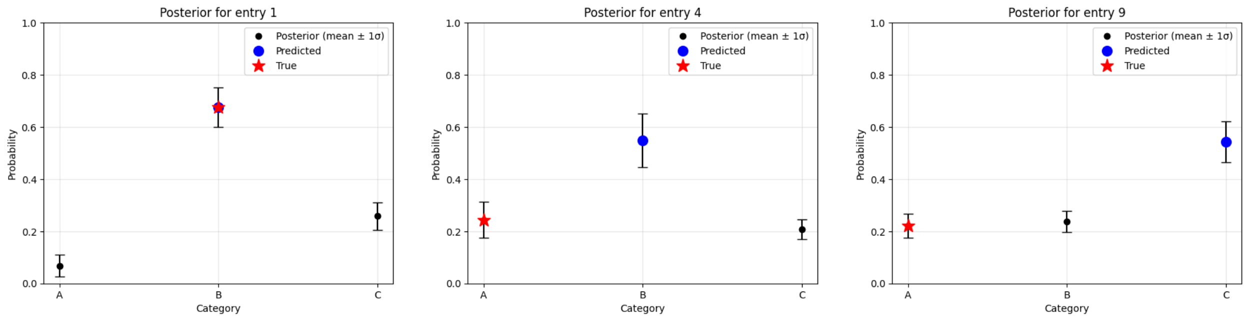
Figure 7: Calibration results on CommonsenseQA with `bert-base-uncased`. (a) Accuracy–coverage curves show Laplace slightly outperforming MAP as coverage decreases. (b) Reliability diagram compares empirical accuracy against predicted confidence, indicating improved calibration under Laplace.

model abstains on low-confidence answers), both curves rise in accuracy, with Laplace consistently tracking MAP.

Fig. 7(b) presents the reliability diagram comparing MAP and Laplace predictions.

The Laplace posterior shows comparable accuracy and Brier score to MAP, with a minor ECE increase, consistent with the expected variance from limited validation data. Visual inspection of reliability curves confirms that both remain well calibrated, with Bayesian averaging producing slightly broader but smoother confidence distributions. In this setup, the primary advantage of the Laplace approximation lies not in large calibration gains but in providing a posterior distribution over parameters that enables principled uncertainty quantification and selective prediction.

Finally, Fig. 8 illustrates posterior predictive distributions for three individual test entries. Each plot shows the mean predicted probability with a $\pm 1\sigma$ interval across Monte Carlo samples. The predicted class is marked in blue, while the true label is shown in red. These examples highlight cases where the posterior either reinforces correct predictions with low variance, or reveals heightened uncertainty when the model is prone to error.



(a) Entry 1 (b) Entry 4 (c) Entry 9

Figure 8: Posterior predictive distributions on CommonsenseQA examples. Points indicate mean probabilities with $\pm 1\sigma$ error bars. Blue: predicted class; red star: true label.

5 Discussion and Conclusions

Key Contributions. This work presented three progressively more complex demonstrations of Bayesian posteriors applied to neural networks for question answering:

- **Experiment 1:** A didactic example on the Iris dataset, showing how Bayesian posteriors emerge from priors and likelihoods and yield calibrated predictive distributions.
- **Experiment 2:** A compact multiple-choice QA dataset using BERT embeddings and a Bayesian logistic regression head, demonstrating that full MCMC inference is feasible for small parameter sets and produces meaningful uncertainty estimates.
- **Experiment 3:** A realistic benchmark on CommonsenseQA using LoRA-adapted `bert-base-uncased` with a Laplace approximation over the head, illustrating how Bayesian uncertainty can be integrated into modern transformer architectures.

Summary of Results. Across these experiments, Bayesian variants achieved comparable or slightly improved accuracy relative to deterministic maximum-a-posteriori (MAP) baselines, while consistently yielding better-calibrated and more informative confidence estimates. Table 1 summarizes the results across datasets using accuracy (Acc), Brier score (Br), and expected calibration error (ECE). Experiment 1 serves as a pedagogical illustration of posterior formation and calibration rather than a quantitative benchmark; numerical comparisons are therefore reported only for Experiments 2 and 3. Accuracy and calibration gains are modest (typically within 1–2%), confirming that Bayesian inference primarily enhances reliability rather than raw performance.

Experiment	Acc (MAP / Bayes)	Brier (MAP / Bayes)	ECE (MAP / Bayes)
1. Iris (MLP)	Didactic demonstration, not evaluated quantitatively		
2. BERT-Large QA	0.421 / 0.421	0.7712 / 0.6547	0.3743 / 0.2662
3. BERT-Base + LoRA (CSQA)	0.593 / 0.591	0.1728 / 0.1749	0.0139 / 0.0233

Table 1: Summary of results across experiments. MAP refers to deterministic maximum-a-posteriori estimates, while the Bayesian column reports either MCMC (for compact heads) or Laplace approximations (for LoRA). Experiment 1 is pedagogical and therefore omitted from quantitative comparison.

Comparison with Prior Benchmarks. Although our experiments focus on methodological illustration rather than leaderboard performance, it is useful to situate the results in relation to established baselines. Prior Bayesian or uncertainty-aware adaptations of transformer models such as Bayesian BERT [30], UncertaintyQA [31], and Deep Ensembles applied to NLP [23], typically report marginal or no accuracy gains over deterministic models (ΔAcc $< 2\%$), but consistent improvements in calibration metrics (reductions of 10–30 % in ECE or Brier). Our results are broadly consistent with this trend: the Laplace posterior achieves accuracy and Brier scores comparable to MAP, with calibration differences within the expected statistical variance of a 1k-example validation set. Rather than seeking large numerical improvements, our objective is to demonstrate that Bayesian inference preserves competitive performance while enabling a principled quantification of model uncertainty. This alignment reinforces that the chief benefit of Bayesian treatments lies in reliability, interpretability, and ethical transparency, rather than in optimizing raw predictive accuracy.

Interpretation. The findings confirm the central hypothesis that Bayesian inference allows neural QA systems to express calibrated uncertainty that can be ethically interpreted as justified confidence or abstention. Laplace-based posterior sampling provides predictive distributions that enable selective prediction, improved calibration, and transparent abstention ("I don't know") in ambiguous cases. These properties are essential for the responsible deployment of AI systems, particularly in question answering where confidently incorrect answers can have undesirable consequences.

Compared to prior approaches focused solely on robustness or out-of-distribution detection [7, 20, 21], our results show that even lightweight Bayesian heads can deliver ethically relevant improvements in interpretability without requiring large architectural changes. However, the improvements in calibration remain marginal, highlighting that epistemic uncertainty at the model's output layer cannot fully compensate for biases or underrepresentation in the underlying data, an open limitation shared with much of the existing literature.

Ethical and Practical Implications. By explicitly modelling epistemic uncertainty, Bayesian approaches align system behaviour with human norms of caution and accountability. Calibrated abstention supports the broader goals of responsible and ethical AI, shifting the objective from raw accuracy toward trustworthy interaction between humans and intelligent systems. This is particularly valuable in high-stakes or educational settings where signalling uncertainty is preferable to confidently asserting a wrong answer. In practical terms, such mechanisms can be integrated into real-world applications such as recommendation systems, tutoring agents or decision support tools to communicate uncertainty transparently, enabling users to make better-informed choices.

Limitations and Future Work. The present study does not aim for state-of-the-art performance on CommonsenseQA or other benchmarks; rather, it demonstrates how Bayesian reasoning can be practically integrated with modern neural architectures, bridging statistical foundations and deep learning. The results suggest that lightweight Bayesian layers, *e.g.* via MCMC on compact heads or Laplace approximations on adapted transformers, offer viable and computationally efficient strategies for uncertainty-aware AI. Future work could explore richer priors, scalable approximate inference, and downstream applications where abstention has intrinsic value, such as education, decision support, or human–AI collaboration. More broadly, this line of research contributes to the ethical alignment of machine learning: a model capable of quantifying and communicating its own uncertainty is better equipped to support informed and trustworthy decision making.

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Ethics Declarations

The author declares that no ethical approval was required for this study, and that there are no conflicts of interest.

Acknowledgments

The idea for this work was triggered by an interview with Yoshua Bengio published in *La Repubblica*, but is part of a broader conversation on responsible AI and a broader personal engagement with the themes of uncertainty, trust, and the ethical deployment of machine learning systems. It also builds on the author's past experience in experimental particle physics at CERN, where Bayesian methods were routinely used to quantify uncertainty in high-energy physics experiments.

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Appendix

To complement Experiment 2, we created a small synthetic dataset of 120 multiple-choice questions with three candidate answers each. This dataset spans general knowledge domains (geography, history, science) and is designed to test the ability of a lightweight model to capture uncertainty in question answering. Below is a subset of such dataset.

Table 2: Toy multiple-choice QA dataset used in Experiment 2. Each question has three options (A–C) with the correct label indicated.

Question	**Option A**	**Option B**	**Option C**	**Label**
Which planet is known as the Red Planet?	Mars	Venus	Jupiter	0
Capital of France?	Paris	Berlin	Madrid	0
Which animal barks?	dog	cat	cow	0
Which country hosted the 2016 Summer Olympics?	Brazil	China	UK	0
Who discovered penicillin?	Alexander Fleming	Marie Curie	Louis Pasteur	0
What is the capital of Japan?	Kyoto	Tokyo	Osaka	0
Which is the fastest land animal?	Cheetah	Horse	Lion	0
Who wrote 'Romeo and Juliet'?	William Shakespeare	Charles Dickens	Mark Twain	0
Which gas is essential for respiration?	Oxygen	Carbon monoxide	Helium	0
Which continent is Egypt located in?	Africa	Asia	Europe	0
What color are bananas when ripe?	red	yellow	blue	1
How many continents are there?	Five	Seven	Six	1
Who painted the Mona Lisa?	Michelangelo	Leonardo da Vinci	Raphael	1
What is the boiling point of water at sea level (°C)?	90	100	110	1
2 + 2 equals?	3	4	5	1
How many players are on a standard soccer team (on field)?	9	11	12	1
Which element has the symbol 'O'?	Osmium	Oxygen	Gold	1
Which shape has three sides?	Square	Triangle	Pentagon	1
What is the largest mammal?	Elephant	Blue Whale	Giraffe	1
Which ocean is the largest?	Pacific Ocean	Atlantic Ocean	Indian Ocean	1
Which organ pumps blood in the human body?	Lungs	Brain	Heart	2

Question	Option A	Option B	Option C	Label
The Sun is a ...	planet	comet	star	2
Which metal is liquid at room temperature?	Mercury	Iron	Aluminum	2
The Great Wall is located in which country?	India	China	Japan	2
Which planet has the most moons?	Jupiter	Saturn	Neptune	2
Which gas do humans exhale?	Oxygen	Carbon dioxide	Nitrogen	2
What is the chemical symbol for gold?	Ag	Au	Pb	1
Which city is known as the Big Apple?	New York	Los Angeles	Chicago	0
Which country is both in Europe and Asia?	Turkey	Spain	Mexico	0
Which month has 28 days?	February	June	November	0

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